CHRISTOPHER WRIGHT, Ph.D.

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RESEARCH AREAS

Firmware and embedded systems security; rehosting, emulation, and digital twins; reverse engineering and vulnerability research; GPU and accelerator security; FPGA and bitstream analysis; binary code similarity; applied and agentic AI for security, including AI red-teaming; compilers, program analysis, and high-performance computing.

EXPERIENCE

GrammaTech, Inc. June 2022 – Present

Principal Scientist and Principal Investigator (2026 – Present) | Senior Scientist III (2025 – 2026) | Senior Scientist II (2023 – 2024) | Lead Software Engineer (2022 – 2023)
Concurrent: Technical Lead, FPGA Security Research

- **Portfolio direction.** Directs concurrent DARPA and government research programs spanning firmware analysis, GPU binary security, and verifiable AI for DoD and IC customers, owning technical direction, staffing and resourcing, and sponsor reporting.
- **Capture.** Won the firmware component discovery and matching effort through the NSTXL consortium for Navy Crane — Phase 1 and Phase 2 delivered, now in Phase 3 transition. Captured the firmware analysis platform as a Direct-to-Phase-II award, then won its Phase II extension, sustaining the program across two consecutive funding actions.
- **Proposals and partnerships.** Leads proposal development and teaming across DARPA, Air Force, and Navy solicitations — including a DARPA Direct-to-Phase-II SBIR as prime on verifiable AI for data analysis — with defense primes, nonprofit research institutes, national laboratories, and universities. Program-manager outreach and profiling across DARPA offices.
- **REAGRIND and WarpGrind.** Multi-vendor GPU binary analysis built on Ghidra P-code: 24+ analysis passes, CWE detectors, bill-of-materials extraction, hardware-validated across NVIDIA SASS sm_60–sm_100; extended to GPU microcontroller firmware for secure-boot analysis. WarpGrind, the companion rewriting toolkit and DARPA ERIS prototype, extends the GTIRB intermediate representation to NVIDIA SASS, applying coverage instrumentation and bounds checking to cubins and reassembling them byte-identically.
- **REAFFIRM, Proteus, FABLE, HALucinator.** Firmware analysis service supporting AI-assisted analysis and SBOM/FBOM generation, delivered under the AFX20 program; hybrid rehosting platform; ML-based component discovery and matching pipeline; technical lead for the HALucinator rehosting and digital-twin platform and its integration into those workflows. Earlier work on cross-architecture function matching over a binary intermediate representation.
- **AI-assisted firmware analysis, made verifiable.** Principal investigator on AI-assisted reverse engineering and firmware analysis, with trustworthy-AI mechanisms built into the workflow: the model plans and narrates but never computes the values it reports, and every reported value is bound to a replayable provenance record.
- **On-premise model infrastructure.** Specified and stood up the GPU inference deployment that grew from a single project server into the company-wide resource, against a CUI and NIST 800-171 compliance boundary.
- **FPGA security research.** Technical lead. Static analysis of FPGA designs recovered from vendor bitstreams; export-controlled.

Sandia National Laboratories May 2019 – July 2022

Senior Scientist and Engineer, R&D Cybersecurity (2021 – 2022) | Graduate Scholar Researcher (2019 – 2021)

- **Intrusion detection for field devices that cannot defend themselves (patented).** Co-invented a system that builds a live digital twin of a deployed RTU or PLC from its own firmware, mirrors production traffic to it, and instruments the twin with defenses the physical hardware cannot carry — heap guards, return-address protection emitted into the emulator's instruction translation, exception-vector detectors, and configuration-integrity checks — detecting unknown attacks by catching attack primitives rather than signatures.
- **VxWorks rehosting and performance.** Built the VxWorks rehosting support layer for HALucinator and replaced interpreted intercept handlers with injected C drivers against native emulated peripherals, taking Ethernet frame handling from seconds to roughly 100–200 ms.
- Reverse engineering and zero-day discovery in source and binary firmware; Ghidra tooling for function matching and execution tracing.

Independent Research 2016 – Present

- **Firmware rehosting at fleet scale.** An agent-driven pipeline producing 219 device rehosts, 176 at verified fidelity across 8 instruction-set families, governed by an M1–M8 evidence ladder in which each rung names the failure mode it excludes and the observation that would refute it. It generates every attack harness itself — no person wrote one — and is steered rather than autonomous by design, with controls that catch it producing confident, well-formed, wrong results. Contributions include a guest-stall control that detects harnesses measuring themselves, and PeriphSynth, hardware model synthesis on local open-weight models with a controlled ablation returning a negative result.
- **Vulnerability discovery.** 42 previously unknown vulnerabilities in commercial embedded and ICS/OT firmware, in coordinated disclosure.
- **AI red-teaming.** Jailbreaking and safeguard bypass, prompt injection and extraction, adversarial inputs against multimodal and agentic systems, and model supply-chain analysis.
- **Binary code similarity.** A calibrated multi-channel consensus method over Ghidra P-code that trains no similarity model of its own, with documented defects in the most-used public benchmark.

- **FPGA hardware Trojan detection.** Static, test-vector-free detection of don't-care-transition Trojans on gate-level netlists reconstructed from bitstreams, using BDD-backed finite-state-machine reachability, with a vendor-neutral reconstruction backend across iCE40, ECP5, 7-series, and UltraScale.
- **Machine learning.** A depth-resolved study of predictive coding against backpropagation, with a companion methodological paper on agent-run pre-registered research.
- **Consulting and advisory.** Selective engagements in architecture design, capability prototyping, technical evaluation, and research collaboration for cybersecurity and embedded-systems teams.
- **Automatic GPU code generation.** A Clang/LLVM source-to-source compiler that identifies C++ code likely to run well on a GPU and translates it to CUDA automatically.

Earlier Research Positions 2015 – 2021

- **Facebook**, Software Engineer Intern, Systems and Infrastructure (2020): polyhedral and numerical analysis over millions of lines of C and C++ for vulnerability discovery, plus source-to-sink taint propagation used by security analysts.
- **Purdue University**, Research Assistant under Milind Kulkarni (2015 – 2021, concurrent with Sandia): firmware emulation and rehosting; compiler and domain-specific language design.
- **Pacific Northwest National Laboratory** (2018) and **Lawrence Livermore National Laboratory** (2016): distributed and parallel algorithms on laboratory supercomputers; LLVM passes recovering the communication structure of MPI programs.

OPEN SOURCE

- **xpu-forge** — Ghidra processor modules for NVIDIA, AMD, Intel, and Apple GPUs. Sole author. NVIDIA SASS (sm_60–sm_120, Pascal through Blackwell, full decode on production cuBLAS, cuSOLVER, and cuRAND libraries), AMD GFX7–GFX11 across 37 targets, Intel Gen9–Xe2, and Apple AGX G13–G16, with SLEIGH grammars, ELF loaders, and custom analyzers. The disassembly backbone of REAGRIND.
- **HALucinator** — contributor and release maintainer for the open-source firmware rehosting framework: VxWorks support, architecture expansion, debugging integration, and the fixes underlying the device fleet above.
- **rehostry** — the device fleet: a per-device repository for each rehosted device, carrying the emulation model, the fidelity evidence and controls, and the framework version it was verified against. Firmware bytes are never committed.

EDUCATION

Ph.D., Electrical and Computer Engineering — Purdue University, 2021. Advisor: Milind Kulkarni. Dissertation: "Emulation for Multiple Instruction Set Architectures."

M.S., Electrical and Computer Engineering — Purdue University

B.S., Computer Engineering — Utah Valley University, 2015, cum laude, minor in Computer Science

SELECTED PUBLICATIONS

- C. Wright, W. Moeglein, S. Bagchi, M. Kulkarni, A. Clements. "Challenges in Firmware Re-Hosting, Emulation, and Analysis." *ACM Computing Surveys* 54(1), Article 5, 2022.
- A. Clements, L. Carpenter, W. Moeglein, C. Wright. "Is Your Firmware Real or Re-Hosted?" Workshop on Binary Analysis Research (BAR), 2021.
- A. Clements, L. Carpenter, W. Moeglein, C. Wright. "Scalable Firmware Re-Hosting." Sandia National Laboratories Technical Report SAND2021-11756, 2021.
- C. Wright, A. Krishnamoorty, M. Kulkarni. "Multi-k de Bruijn Graph Based Genome Assembly." Algorithms for Computational Biology (AlCoB), 2019.

UNDER SUBMISSION

- C. Wright. Sole-authored paper on a falsifiable protocol for automated firmware rehosting at fleet scale. Network and Distributed System Security Symposium (NDSS) 2027.
- C. Wright. Sole-authored paper on calibrated multi-channel consensus for deployment-scale binary code similarity. Network and Distributed System Security Symposium (NDSS) 2027.

PATENTS

- "System and Methods for Intrusion Detection and Prevention Using Firmware Re-Hosting." Sandia National Laboratories, under U.S. Department of Energy / NNSA Contract DE-NA0003525.
- "System and Method for Static GPU Binary Instrumentation via Processor Specification Round-Trip with Shadow-Slot Exfiltration." Provisional application filed August 2026. Sole inventor.

TEACHING, SERVICE, AND AFFILIATIONS

- Instructor, Purdue University ECE 368, Data Structures and Algorithms (evaluations 4.5 and 4.7 of 5); teaching assistant for compiler construction and data structures.
- Member, IEEE, ACM, and Eta Kappa Nu. Security clearance: details available on request.